Programme-Driven Demand Shocks in Food Supply Chains: System Dynamics and Machine-Learning Scenario Discovery for Indonesia's Free Nutritious Meals Programme

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Abstract

Indonesia's Free Nutritious Meals programme (Makan Bergizi Gratis, MBG) has created a large, fast-growing and price-inelastic public demand for eggs and broiler meat, and whether it drives food price increases is disputed between government agencies. This paper develops a system dynamics model of the egg and broiler supply chains under MBG that represents kitchen roll-out and suspensions, menu and school-holiday rules, biological production delays, farmer investment and price formation. The model is calibrated to national retail prices and programme statistics and validated with structural and behavioural tests. It is run as an exploratory ensemble of 5,000 scenarios, each simulated with and without the programme so that programme-attributable price effects are separated from seasonal and external shocks, and machine-learning scenario discovery (classifiers, SHAP and PRIM) is used to identify the conditions that produce price spikes. Under current policy, the programme raises egg or chicken prices by more than 10 percentage points in 86.0% of scenarios. Spike risk is driven mainly by chicken quantity per beneficiary and egg servings per week after June 2026; a one-at-a-time sensitivity analysis identifies the same main driver, and scenario discovery adds the probability of a spike and the threshold beyond which spikes become likely. Forward contracting (P3) is the most effective single policy, lowering spike probability to 65.7%, and combining cooperative sourcing, forward contracts and menu flexibility lowers it to 38.5% while reducing protein cost per portion. Indicators observable early in the roll-out anticipate spikes only moderately (ROC AUC 0.80, with accuracy no better than always predicting a spike), which favours designing procurement in advance over reacting to prices.

Keywords: system dynamics; scenario discovery; machine learning; food supply chain; demand shock; school meals programme; public procurement; Indonesia

1 Introduction

1.1 Motivation

Large public food programmes do more than feed their beneficiaries: they also change the markets they buy from. Indonesia's Free Nutritious Meals programme (Makan Bergizi Gratis, MBG), launched on 6 January 2025, is one of the largest public food procurement programmes in the world. By the end of 2025 the National Nutrition Agency (Badan Gizi Nasional, BGN) operated 19,188 meal-preparation units (Satuan Pelayanan Pemenuhan Gizi, SPPG) serving 55.1 million beneficiaries [1], rising to 21,897 units and 65 million beneficiaries by February 2026 [2]. The programme budget rose from Rp 171 trillion in 2025 to a proposed Rp 335 trillion in 2026 [3].

Eggs and broiler meat are core protein items on MBG menus. Unlike household demand, programme demand is set by menus, kitchen numbers and the school calendar rather than by price, and it grows as fast as new kitchens open. Supply adjusts slowly: pullets take about 18 weeks to reach point of lay, and farmers expand flocks on the basis of expected prices and margins. A fast, price-inelastic increase in demand meeting a slow, biologically delayed supply response is a classic setting for price overshoot and commodity cycles [4, 5].

The first two years of the programme show this tension, and its interpretation is contested. Statistics Indonesia (BPS) attributed part of the October 2025 inflation in eggs and chicken meat to demand from MBG kitchens [6], whereas the Ministry of Agriculture described prices as stable and supply as sufficient [7]. In November 2025 the President asked kitchens to replace chicken eggs temporarily with other proteins as prices rose [8]. In June 2026, when farm-gate egg prices fell below the government reference price, BGN and farmer groups agreed to serve eggs at least three times a week [9]; when meals paused for the school holiday the Minister of Agriculture asked companies to absorb farmers' eggs and chicken [10], and when schools reopened he attributed the following price rise to MBG demand [11]. Menus, serving days and holiday rules have therefore become de facto market-stabilisation instruments, adjusted reactively.

These events raise questions that statistical analysis of short price series cannot settle: how much of the price movement is caused by the programme rather than by seasonal or external shocks; under which conditions programme demand produces a severe price spike; and which policies, from the speed of roll-out and the sourcing structure to advance contracting with farmers and flexible menus, reduce that risk. Answering them requires a model that captures feedback and delays across the supply chain, and an analysis that explores many plausible futures rather than a single forecast.

1.2 Novelty and Contributions

1. A system dynamics model of Indonesia's egg and broiler supply chains under a national school-meals programme, calibrated to public price and programme data and including the programme's menu and school-holiday rules, with a counterfactual design that separates programme effects from seasonal and external shocks (Sections 3 and 4).
2. A combination of system dynamics and machine-learning scenario discovery (a classifier benchmark, SHAP and PRIM) that yields interpretable spike conditions, compared with one-at-a-time sensitivity analysis (Section 5.3).
3. An early-warning analysis testing whether indicators observable 13 weeks after programme launch predict later price spikes (Section 5.4).
4. A paired statistical comparison of roll-out and procurement policies on 5,000 common scenarios (Section 5.5).

1.3 Paper Organization

Section 2 reviews related work. Section 3 describes the research questions, model and analysis, and Section 4 the data. Section 5 presents the results, Section 6 discusses them, Section 7 states limitations and directions for future work, and Section 8 concludes.

2 Related Work

2.1 Public Food Procurement and School Meals

School meals programmes are usually evaluated for their effects on children's nutrition and learning, but large programmes also act on food markets. Handbury and Moshary [12] show that the United States National School Lunch Program affects retail food markets beyond the children it serves, so that a public meals programme has consequences for households that do not participate in it. Home-grown school feeding links meals to local smallholder procurement in the expectation of raising farm demand and incomes; a cluster-randomised evaluation in Ghana, however, found no strong evidence that the programme changed local market structure or farm incomes, and argued that effects depend on whether local markets are integrated with wider markets and on the scale of procurement [13].

Research on MBG is recent and mainly descriptive. A study of Tangerang Selatan maps the programme's food supply chain from ingredient sources through procurement channels and kitchens to beneficiaries, and finds that local small enterprises take part largely through partners and cooperatives acting as procurement channels [14]. In Central Java, sourcing from local producers had not been achieved because of constraints in quality, quantity, continuity and price, and supply still relied on inter-regional market mechanisms rather than local contracts [15]. These studies describe procurement structures but do not quantify how programme demand moves prices over time.

2.2 Demand Shocks, Biological Lags and Commodity Cycles

Livestock markets are prone to cycles because production capacity responds to prices only after a biological delay. Meadows [4] showed that commodity production cycles arise endogenously when producers invest on the basis of current prices and ignore capacity already in the pipeline, a mechanism generalised by Sterman [5] as supply-line underestimation. Mitchell, Thompson and Malone [16] show the empirical importance of such lags for eggs: highly pathogenic avian influenza raised weekly United States retail egg prices by 5.3% on average, and models that ignore the time needed to replace a depopulated flock underestimate the impact. The shocks studied in this literature are disease, feed and seasonal shocks, not demand shocks created by public programmes.

2.3 System Dynamics in Poultry Supply Chains

System dynamics has been used to explain poultry market behaviour. Haryana et al. [17] build a model of Indonesia's broiler supply system, linking grandparent and parent stock, day-old-chick allocation, integrator and independent grow-out, slaughter and cold storage, and show that recurring live-bird price instability arises from the structure of the system; they conclude that stabilisation requires coordinated management of breeding capacity, hatchery operations and downstream buffering. Aboah and Enahoro [18] model backyard poultry farming in Ghana over a 52-week horizon and show that the drivers of the system change within the year. Neither study considers a public programme as a source of demand, and both analyse a small number of scenarios.

2.4 Scenario Discovery and Machine Learning on Simulation Models

Scenario discovery applies statistical learning to large ensembles of simulation runs to find the combinations of uncertain inputs that lead to outcomes of interest [19]. The Patient Rule Induction Method (PRIM) [20] finds a box in the input space with a high share of such outcomes and reports the trade-off between coverage and density; later work extended the approach to heterogeneous uncertainties [21] and to behaviour patterns over time [22], and the EMA Workbench made it widely available [23]. Tree ensembles such as random forests [24] combined with SHAP values [25] offer a complementary, model-agnostic way to rank and interpret drivers. These methods have been applied mainly to energy, water and climate policy, and rarely to food supply chains or to public programmes in developing economies.

2.5 Research Gap

Table 1 positions this study. Existing work either measures market effects of meals programmes without modelling supply dynamics, or models poultry supply dynamics without programme demand, and none uses scenario discovery. This study combines the three.

3 Methodology

3.1 Research Questions

RQ1: How does the MBG roll-out propagate through Indonesia's egg and broiler supply chains to market prices and household consumption over time? **RQ2:** Under which combinations of uncertain conditions does the programme cause severe programme-attributable price spikes? **RQ3:** Which roll-out and procurement policies reduce spike risk while maintaining programme coverage and cost?

3.2 Research Design

The study combines system dynamics (SD) simulation with machine-learning scenario discovery (Figure 1). An SD model of the egg and broiler supply chains under MBG was built in Vensim, calibrated to public data and validated with structural and behavioural tests [26]. It was then run as an exploratory ensemble over uncertain parameters and alternative policies using PySD [27]. Classifiers, SHAP values [25] and PRIM [19, 20], implemented with the EMA Workbench [23], were used to identify the conditions under which the programme produces severe price increases, and policies were compared on a common set of scenarios with paired non-parametric tests.

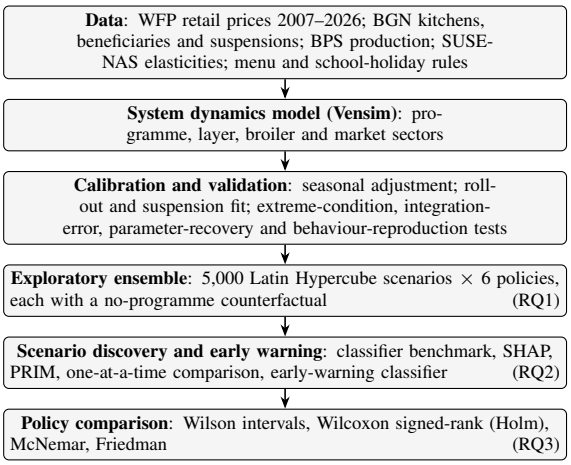


Figure 1: Research framework

3.3 Model Structure

The model has four interacting sectors simulated weekly from January 2024 to December 2029 (312 weeks, Euler integration with a time step of 0.03125 week). The step is one-sixteenth of the shortest time constant, the half-week minimum delivery time that limits sales when demand jumps as schools reopen, and school-holiday dates are rounded to the nearest 1/16 week so that both time steps in the integration tests switch meals on and off at the same moments. Figure 2 shows the main feedback loops, Figure 3 the stock-and-flow structure, and Appendix A lists all equations.

Programme sector. Planned kitchens (SPPG) follow a logistic roll-out whose final size, midpoint and steepness were fitted to BGN's reported counts of operational kitchens. Kitchens open after an opening delay and can be suspended; the weekly suspension fraction rises when the opening rate exceeds a sustainable opening capacity, representing strain on supervision, staff and suppliers during rapid scaling, and suspended kitchens reopen after a reinstatement time. Beneficiaries served equal operational kitchens times beneficiaries per kitchen, which follows the reported ratio up to September 2026 and then moves to an uncertain future level. Programme demand equals beneficiaries served times the menu quantity per beneficiary times a school-calendar factor: meals pause during school holidays, except for pregnant and breastfeeding mothers and young children until BGN stopped holiday distribution to all groups in June 2026 [28]. Egg demand per beneficiary is the serving size times the number of egg servings per week, which rises from two to at least three on 8 June 2026 [9].

Egg (layer) sector. Pullets are placed and mature into laying hens through a third-order rearing delay. Hens leave the flock through culling and disease outbreaks. Farmers place pullets to replace expected losses and to close the gap between desired and actual flock size, with a partial correction for pullets already in rearing (the supply-line weight [5]). Desired flock size responds to the expected margin, formed by smoothing the market price over an expectation delay (balancing loop B1). Placements are limited by hatchery capacity.

Broiler sector. Day-old chicks are placed and harvested after grow-out. Placements follow smoothed demand, an inventory correction and the expected margin, limited by day-old-chick supply.

Market and price formation. For each commodity, market inventory accumulates production and is depleted by sales and spoilage; programme kitchens are served before households when stock is short. Household demand responds to price through a constant own-price elasticity (balancing loop B2). Price adjusts toward an indicated price that rises when inventory coverage is below normal and is pulled toward unit cost plus a normal margin, bounded by a maximum relative price that represents imports, substitutes and market operations. Unit cost depends on the feed price, which rises with the total flock (balancing loop B3) and with exogenous feed-price shocks. Under menu flexibility (policy P4), kitchens substitute other proteins when prices are high, closing a further balancing loop (B4).

Table 1: Positioning of this study relative to prior work

Study	Context	Method	Programme demand	Dynamic feedback	Scenario discovery
Handbury & Moshary [12]	USA, school lunch	Econometric	Yes	No	No
Gelli et al. [13]	Ghana, school feeding	Randomised trial	Yes	No	No
IPB study [14]; Ambarwati & Kurniawati [15]	Indonesia, MBG	Descriptive	Yes	No	No
Mitchell et al. [16]	USA, eggs, avian influenza	Econometric with lags	No	Partly	No
Haryana et al. [17]	Indonesia, broilers	System dynamics	No	Yes	No
Aboah & Enahoro [18]	Ghana, backyard poultry	System dynamics	No	Yes	No
Bryant & Lempert [19]; Kwakkel [23]	Methods	Scenario discovery	No	–	Yes
This study	Indonesia, MBG, eggs and broilers	SD + ML scenario discovery	Yes	Yes	Yes

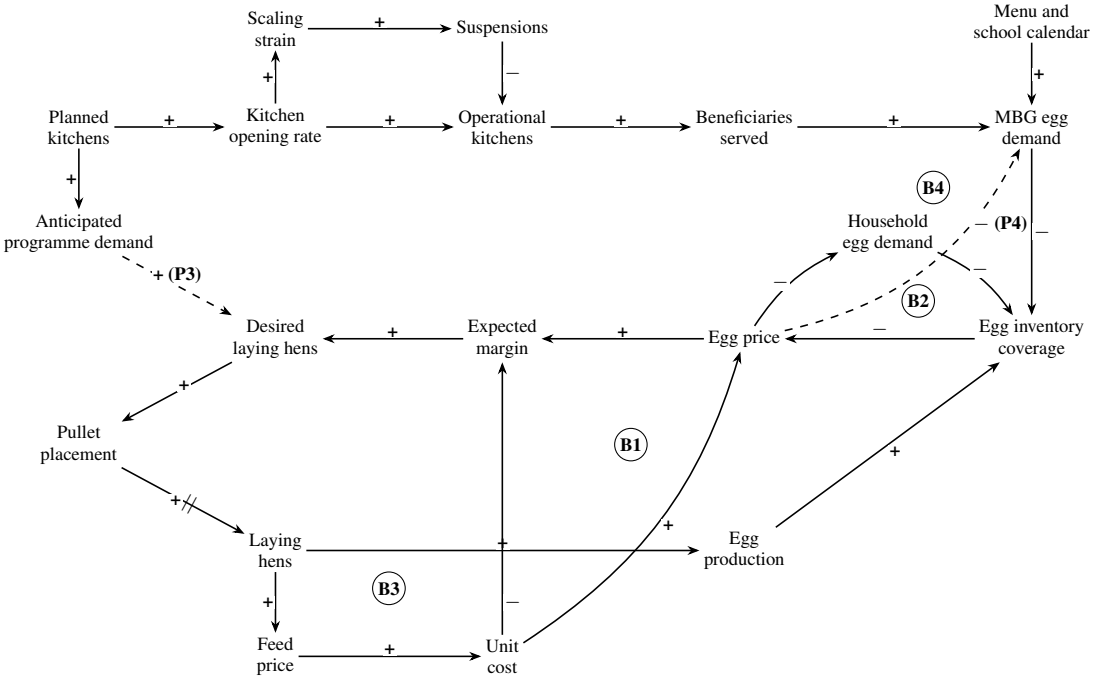


Figure 2: Causal loop diagram of the egg chain (the broiler chain is analogous). Dashed links act only under policies P3 (forward contracts) and P4 (menu flexibility); || marks the rearing delay

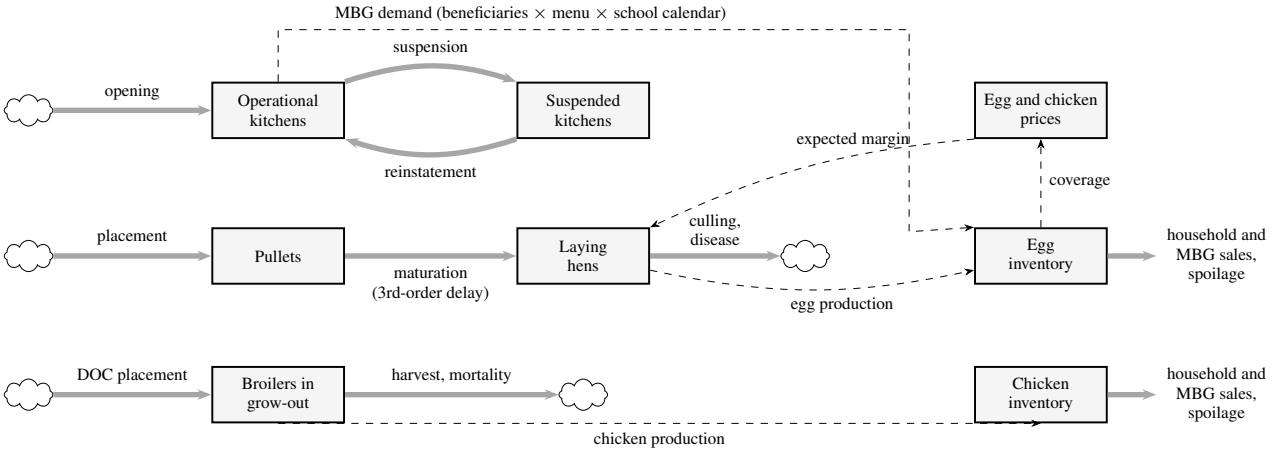


Figure 3: Simplified stock-and-flow structure (stocks as boxes, flows as thick arrows, information links and production rates computed from stocks dashed). Expected prices and the feed price are omitted for clarity

3.4 Policies

P1 slows the roll-out so that 10–90% of kitchens open over about 13 months instead of about 7. P2 reflects the finding that MBG procurement runs largely through intermediaries [14, 15]. P4 formalises the reactive menu changes of November 2025 [8] as a rule.

Table 2: Policy levers

	Policy	Implementation in the model
P0	Baseline	Current roll-out; cooperative sourcing share 0.1; no forward contracts; fixed menu
P1	Phased roll-out	Roll-out steepness multiplied by 1.75 (a slower S-curve)
P2	Cooperative sourcing	Direct purchase share from farmer groups 0.6 (lower intermediary mark-up)
P3	Forward contracts	80% of future programme demand contracted 20 weeks ahead, so farmers can expand flocks before the demand arrives
P4	Menu flexibility	Up to 50% substitution of other proteins when the relative price exceeds 1.15
P5	Combined	P2 + P3 + P4

3.5 Validation

Following Barlas [26] and Sterman [5], the model was subjected to dimensional consistency checks; six extreme-condition tests (no programme, ten-fold programme demand, loss of all laying hens, no hatchery expansion, doubled feed price and near-instant roll-out); an integration-error test comparing the baseline run at two time steps; an ensemble robustness test repeating all ensemble scenarios for four policies at both time steps; a parameter-recovery test applying the calibration procedure to synthetic data with known parameters; and behaviour reproduction against historical prices, assessed with MAPE, R^2 and Theil inequality statistics [29].

3.6 Exploratory Ensemble and Outcome Indicators

5,000 scenarios were drawn by Latin Hypercube sampling [30] over 23 uncertain parameters (Table 3). Each policy was run on the same scenarios (common random numbers), giving a paired design, and each scenario was also run without the programme. To make the ensemble feasible at the small time step, blocks of scenarios were integrated together as arrays with the same Euler update as PySD; before use, these runs were checked against ordinary PySD runs (21 runs, maximum relative difference 8e-15). Price effects were measured as the difference between the two runs, so that movements caused by exogenous shocks are not attributed to the programme. Outcome indicators were the peak programme-attributable egg and chicken price rise, weeks above the threshold, mean household consumption loss, farmers' margin volatility, protein cost per portion and programme coverage in the final year. A run was labelled a spike if the programme-attributable price rise in either commodity exceeded 10 percentage points of the baseline price. The threshold is well above the largest seasonal premium in the pre-programme data (chicken prices 5.8% above trend before Idul Fitri; Section 4.1), so a spike is a rise beyond normal seasonal variation; for eggs it corresponds to about Rp 2,814 per kilogram.

Table 3: Uncertain parameters and ranges

Parameter	Low	High	Basis
Rollout midpoint	86	100	Fitted value $\pm$ about 7 weeks
Rollout steepness	5	12	Fitted value $\pm$ about 50%
Target kitchens	2.4e+04	3.2e+04	Fitted 29,000; BGN targets up to 32,000
Future beneficiaries per kitchen	1,900	2,900	Sep. 2026 ratio to earlier ratio; cap 2,500–3,000 [31]
Egg serving size	0.05	0.065	One egg, 50–65 g
Egg servings after menu change	2	4	Two to four per week [9]
Chicken per beneficiary	0.08	0.26	Conflicting official estimates [32]
Price expectation delay	4	16	Not identifiable (Table 6)
Investment sensitivity	0.5	3	Around calibrated value
Supply line weight	0.1	1	Full range [5]
Hatchery capacity factor	1.1	2	Spare hatchery capacity
DOC capacity factor	1.05	1.4	Day-old-chick oversupply 1.05–1.4
Household egg elasticity	-1	-0.5	QUAIDS estimates [33]
Household chicken elasticity	-1.8	-1.2	QUAIDS estimates [33]
Price sensitivity to coverage	0.5	3	Not identifiable (Table 6)
Feed shock magnitude	0	0.5	Up to +50% feed price
Outbreak loss fraction	0	0.3	Up to 30% of hens lost
Base suspension fraction	0.002	0.008	Fitted 0.0042; 0.5–2 times
Strain sensitivity	0	5	No data; wide range
Kitchen opening delay	2	8	No data; 2–8 weeks
Sustainable opening capacity	150	700	No data; wide range
Intermediary margin	0.1	0.3	BPS trade margins for eggs 13–27%
Maximum relative price	2	4	Imports, substitutes, market operations

3.7 Scenario Discovery and Early Warning

Using baseline-policy runs, four classifiers (logistic regression, a depth-3 decision tree, a random forest [24] and gradient boosting) were trained to predict spikes from the uncertain parameters and evaluated by stratified five-fold cross-validation. The decision tree provides readable rules; SHAP values on the random forest quantify each parameter's contribution and direction; PRIM identifies a region of the uncertainty space with a high spike density: the box with the highest coverage whose density is at least halfway between the ensemble spike share and one (a fixed target such as 0.8 would select the whole space when spikes are already that common), reporting dimensions restricted by at least 10% of their range. The results were compared with a one-at-a-time sensitivity analysis. An early-warning classifier was trained on indicators observable 13 weeks after programme start: relative prices, inventory coverage, growth in pullet placements, the kitchen opening rate, the share of suspended kitchens and the feed price.

3.8 Policy Comparison

For each policy, spike probability is reported with Wilson 95% confidence intervals [34], together with the median and 90th percentile of each outcome. Each policy is compared with P0 on the same scenarios with the Wilcoxon

signed-rank test [35], Holm-corrected [36], with the matched-pairs rank-biserial correlation as effect size [37]; spike occurrence is compared with an exact McNemar test [38], and all policies jointly with the Friedman test [39].

4 Data

Table 4 summarises the data used to build, calibrate and validate the model. Appendix B lists every parameter with its value and source.

Table 4: Overview of data sources

Data	Source	Use in the model	Period
Retail prices of eggs and chicken	WFP market prices [40]	Calibration; baseline prices; seasonality	2007–2026
Operational SPPG	BGN [1, 41] and press reports [42]	Roll-out fit	Jan. 2025–Sep. 2026
Beneficiaries	BGN [1, 2, 41]; Kemenkeu [42]	Beneficiaries per kitchen	Jan. 2025–Sep. 2026
Kitchen suspensions	BGN	Suspension hazard	to May 2026
Egg and chicken production	BPS [43]	Initial production	2024
Household price elasticities	QUAIDS on SUSENAS [33]	Household demand	2021
Menu and holiday rules	BGN statements [9, 28, 44]	Programme demand	2025–2026
Biological and cost parameters	Breed standard [45]; Indonesian farm and trade statistics	Production and unit cost	various

4.1 Prices and Seasonality

Monthly retail prices of chicken eggs and chicken meat were taken from the World Food Programme's Indonesia food price dataset [40], which covers about 200 markets. Because the set of reporting markets varies, each market's price was expressed relative to its own 2024 mean, and the national index for each month is the median of these relatives. Months with fewer than 100 reporting markets, and the incomplete latest month, were excluded, leaving 29 monthly observations from January 2024 to October 2026. Baseline prices were set to the 2024 mean of monthly national medians (Rp 28,140/kg for eggs and Rp 35,704/kg for chicken).

The model contains no seasonal household demand, so seasonality was removed before calibration. Using the WFP national series for 2007–2023 (March 2020 to December 2021 excluded), the log ratio of price to its centred 12-month moving average was regressed on calendar-month dummies and on the shares of the 30 days before and after Idul Fitri falling in each month, with HAC standard errors [46]. Before Idul Fitri, egg prices are 3.6% above trend ($p = 0.002$) and chicken prices 5.8% above trend ($p < 0.001$). The fitted seasonal factors were divided out of the 2024–2026 series.

4.2 Programme Data

Operational SPPG counts were compiled for 13 dates from January 2025 to September 2026 from BGN releases and press reports of BGN statements [1,41,42]. Reported beneficiaries per operational kitchen were about 2,700–3,200 until February 2026 but about 2,000 in September 2026 (53–57 million beneficiaries in 27,485 kitchens) [42]; the model follows the observed ratio and treats the later level as uncertain, bounded by BGN's cap of 2,500 portions per kitchen (3,000 with certified cooks) [31]. Suspension parameters were fitted to BGN's report that 8,182 SPPG had been suspended at some point and 2,213 remained inactive at the end of May 2026, giving a weekly suspension hazard of about 0.4% and a mean reinstatement time of about 15 weeks.

Egg servings follow BGN's planning assumption of two egg dishes a week [44] until 8 June 2026 and at least three afterwards [9]. No menu specification for chicken has been published, and official estimates of programme chicken demand conflict, from about 0.04 to 0.26 kg per beneficiary per week [32]; chicken quantity is therefore treated as a deep uncertainty. Initial production was set from BPS 2024 figures: 6,342,705 t of eggs and 3,835,917 t of broiler meat (preliminary) [43]. Household own-price elasticities were set from QUAIDS estimates on SUSENAS 2021 data (-0.669 for eggs and -1.581 for poultry) [33]. Layer parameters (point of lay at about 18 weeks, laying period of about 65 weeks, hen-day production of about 84.5%) follow the breed standard [45] and Indonesian farm practice; feed is 60–70% of production cost and trader mark-ups follow BPS trade and transport margins for eggs (13–27% in 2017–2021).

4.3 Calibration

The investment sensitivities of layer and broiler farmers were estimated jointly on the seasonally adjusted egg and chicken series by minimising the sum of squared errors (Nelder–Mead, four starting points). Two further parameters, the price-expectation delay and the sensitivity of price to inventory coverage, are not identifiable from these data: the fit error changes little across wide ranges of either (Table 6). They were fixed at 8 weeks and 1.0, values that do not depend on the data, and explored over wide ranges in the ensemble.

5 Results

5.1 Validation and Calibration

All six extreme-condition tests passed (Table 5). The integration-error test gave a maximum difference of 0.63% of each variable's range between time steps of 0.03125 and 0.015625 week (within the 1% criterion), and 99.7% of ensemble runs received the same spike classification at both time steps. The parameter-recovery test

recovered all parameters within 1.7% of their true values. The calibrated layer investment sensitivity is 2.80; the broiler investment sensitivity is 0.000, at the lower bound of zero.

On the seasonally adjusted series the model reproduces egg prices with a MAPE of 3.1% and an R^2 of 0.45, and chicken prices with a MAPE of 4.1% and an R^2 of 0.57 (Table 7, Figure 4). The bias proportion of the error (U_M) is 0.026 for eggs and 0.000 for chicken, and most of the error is unsystematic (U_C = 0.67 and 0.61).

5.2 Baseline Dynamics

Figure 5 compares the calibrated baseline with the same run without the programme. The programme-attributable egg price rise peaks at 11.2 percentage points in April 2029 and the chicken price rise at 12.8 percentage points in April 2026. The egg price exceeds the counterfactual by more than 10 points for 6 weeks and the chicken price for 9 weeks. Average household consumption over the programme period is 1.0% lower for eggs and 5.8% lower for chicken than without the programme. School holidays produce short, sharp drops in programme demand that are visible in the price paths.

5.3 Conditions Leading to Programme-Attributable Spikes

Under the baseline policy, 86.0% of scenarios produce a spike (95% CI 85.0%–86.9%). The spike outcome is predictable from the uncertain inputs: gradient boosting reaches a cross-validated ROC AUC of 0.967, against 0.959 for logistic regression and 0.870 for a depth-3 tree (Table 8). The small gap between logistic regression and the tree ensembles shows that interactions between inputs add little: spike risk is largely additive in the uncertain parameters.

SHAP values (Figure 6) rank chicken quantity per beneficiary, egg servings per week after June 2026 and size of a disease outbreak as the main drivers. The PRIM box (Table 9) covers 80% of all spikes with a spike density of 94% (target 93%), against 86% in the whole ensemble, and restricts chicken quantity per beneficiary (to 0.128–0.26). The one-at-a-time analysis (Figure 9) ranks chicken quantity per beneficiary, size of a disease outbreak, household chicken price elasticity highest; its ranking correlates moderately with the SHAP ranking (Spearman ρ = 0.66). Both identify chicken quantity per beneficiary as the main driver. The rankings differ for second-tier drivers: egg servings per week after June 2026 ranks second by SHAP but sixth one at a time, because one-at-a-time variation measures each effect only with the other inputs at their mid-points, whereas SHAP averages over the whole ensemble.

5.4 Early-Warning Indicators

Indicators observed 13 weeks after programme start predict later spikes only moderately (Table 10): the random forest reaches a ROC AUC of 0.796 and a depth-2 tree 0.613. The first split of the tree uses the chicken coverage (Figure 10). At that point programme demand is still small relative to supply, so the market signals that would warn of a later spike have not yet formed. Because spikes occur in 86% of scenarios, accuracy (86% at best) is no better than always predicting a spike, and every leaf of the depth-2 tree predicts a spike; only the ranking of scenarios by risk, measured by the ROC AUC, carries information.

5.5 Policy Comparison

Table 11 and Figure 11 compare the policies. Against 86.0% under P0, the spike probability is P1 (phasing the roll-out) 85.1%, removing 161 spikes and adding 120; P2 (cooperative sourcing) 86.0%, removing 0 spikes and adding 0; P3 (forward contracting) 65.7%, removing 1,137 spikes and adding 122; P4 (menu flexibility) 78.3%, removing 383 spikes and adding 0; P5 (the combined policy) 38.5%, removing 2,381 spikes and adding 9 (Table 13). Median protein cost per portion is Rp 2,404 under P0, Rp 2,202 under P2 and Rp 1,973 under P5. Programme coverage in the final year is the same under all policies. After Holm correction, P2 does not differ significantly from P0 on peak egg price, peak chicken price or household egg loss. The Friedman test rejects equal outcomes across policies for every indicator tested ($p < 0.001$).

Table 14 repeats the comparison under other spike definitions: lower and higher thresholds, and a sustained definition requiring the rise to stay above 10 points for several weeks. P5 and P3 are the two best policies under 4 of the 6 definitions, and most baseline spikes are sustained: 80.5% of scenarios stay above 10 points for at least four weeks. For severe spikes (above 20 points) the ranking of single policies changes: menu flexibility lowers the probability to 3.8%, against 19.6% for forward contracting, because substitution caps how far kitchens keep buying when prices are high.

6 Discussion

6.1 Why the System Behaves This Way

Programme demand is set by kitchen numbers, menus and the school calendar, not by price, and it grew from almost nothing to tens of millions of meals within a year. Laying capacity cannot follow at that speed: a pullet placed in response to higher prices starts laying about 18 weeks later, and farmers form price expectations with a delay. In the meantime the market clears through price, and the balancing loop that acts fastest is household demand (B2): households buy fewer eggs and less chicken, which is why the programme-attributable price rise in the baseline coincides with household consumption 1.0% (eggs) and 5.8% (chicken) lower than without the programme. The programme's nutritional gain for beneficiaries is therefore partly offset by a loss for non-beneficiary households, a distributional effect that does not appear in programme statistics.

Table 5: Validation tests

Test	Pass criterion	Observed	Result
A1 No programme -> system stays in equilibrium	max |relative egg price - 1| < 1e-6	0.00e+00	Pass
A2 10x programme demand -> price up, household consumption down	peak price > 1.2 and household ratio < 0.9; finite, no negative stocks	peak=2.68, hh_min=0.52	Pass
A3 Outbreak kills 100% of hens -> production collapses, price spikes, no negatives	min hens < 10% of initial; 1.5 < peak price <= ceiling; no negative stocks	min hens=3,358,814, peak=3.00	Pass
A4 No hatchery expansion possible -> higher prices than baseline	mean price (weeks 100-312) above baseline run	1.118 vs 1.015	Pass
A5 Feed price doubles -> egg price rises during shock	price at week 140 above baseline run	1.090 vs 1.007	Pass
A6 Near-instant roll-out (steepness 0.5 wk) -> more suspensions than baseline (scaling strain)	peak suspended kitchens above baseline run	5,410 vs 3,177	Pass
B1 Integration error: TIME STEP 0.03125 vs 0.015625	max |difference| < 1% of each variable's range	0.6328%	Pass
B2 Ensemble robustness: TIME STEP 0.03125 vs 0.015625 (5000 scenarios x 4 policies)	same spike classification in >= 98% of runs	99.7% agree; median |peak diff| 0.0003	Pass

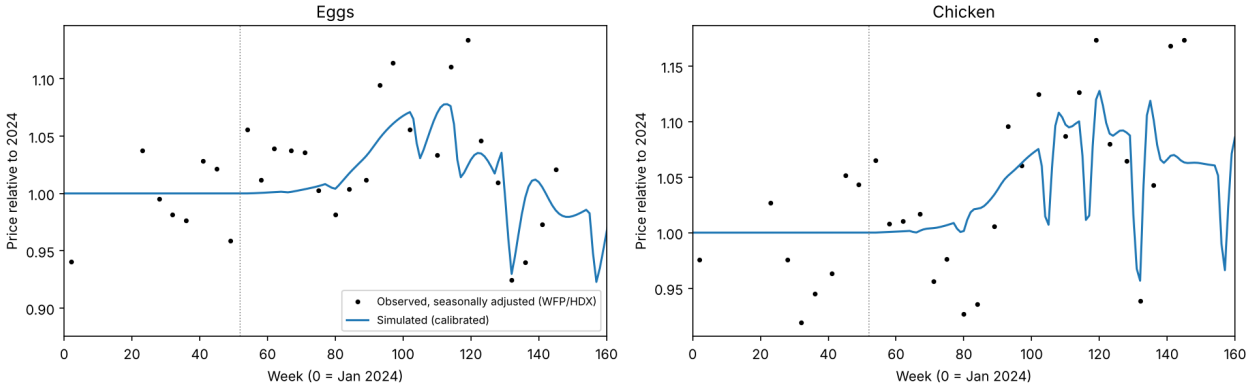


Figure 4: Simulated and observed (seasonally adjusted) relative prices

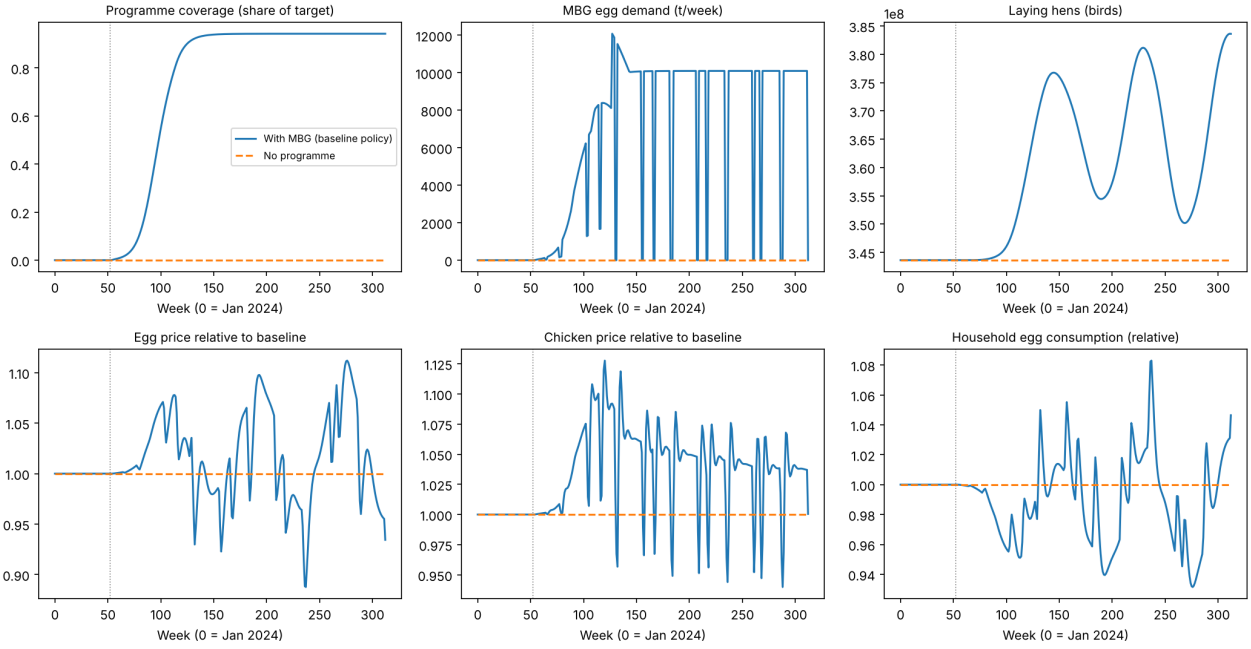


Figure 5: Baseline dynamics with and without the programme

Once new flocks come into lay, the supply loop (B1) overshoots: in the baseline the egg price falls 6.6 percentage points below the counterfactual by December 2029. This is the commodity-cycle mechanism described by Meadows [4] and Sterman [5], triggered here by a demand shock rather than by a supply shock. It matches the sequence observed in 2026, when farm-gate egg prices fell below the government reference price and BGN raised egg servings to absorb the surplus [9].

School holidays act as a second, recurring shock. When meals stop for two to three weeks, programme demand drops at once, but eggs keep coming from flocks sized for term-time demand, so inventories rise and prices fall; when schools reopen, demand returns at full strength. The June 2026 request by the Minister of Agriculture that companies absorb farmers' eggs and chicken during the

Table 6: Identifiability profile: fit error when one fixed parameter is varied

Fixed parameter	Value	SSE
Price sensitivity to coverage	0.5	0.00460
Price sensitivity to coverage	1.0	0.00402
Price sensitivity to coverage	2.0	0.00391
Price sensitivity to coverage	3.0	0.00394
Price sensitivity to coverage	5.0	0.00393
Price sensitivity to coverage	10.0	0.00396
Price expectation delay	2.0	0.00424
Price expectation delay	4.0	0.00412
Price expectation delay	8.0	0.00402
Price expectation delay	16.0	0.00408
Price expectation delay	30.0	0.00448

Table 7: Behaviour reproduction against historical prices

Series	MAPE	R^2	U_M	U_S	U_C
egg (seasonally adjusted)	0.031	0.446	0.026	0.300	0.674
chicken (seasonally adjusted)	0.041	0.573	0.000	0.386	0.614
egg (raw, sim x seasonal factor)	0.031	0.261	0.022	0.044	0.934
chicken (raw, sim x seasonal factor)	0.041	0.441	0.000	0.234	0.766

Table 8: Spike classifiers (five-fold cross-validation, mean $\pm$ SD)

Classifier	Accuracy	F1	ROC AUC
Logistic regression	0.921 ± 0.009	0.955 ± 0.005	0.959 ± 0.008
CART (depth 3)	0.877 ± 0.004	0.930 ± 0.003	0.870 ± 0.010
Random forest	0.903 ± 0.009	0.946 ± 0.005	0.959 ± 0.005
Gradient boosting	0.929 ± 0.010	0.960 ± 0.005	0.967 ± 0.007

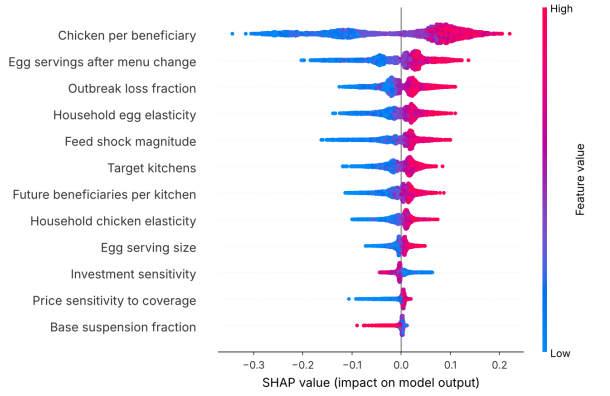


Figure 6: SHAP summary for the random-forest spike classifier

Table 9: PRIM box (coverage 0.80, density 0.94, mass 0.73; density target 0.93)

Parameter	Box low	Box high	Full range	Removed
Chicken quantity per beneficiary	0.128	0.26	0.08–0.26	27%

Table 10: Early-warning classifiers using indicators observed 13 weeks after programme start

Classifier	Accuracy	F1	ROC AUC
CART (depth 2)	0.860	0.924	0.613
Random forest	0.861	0.924	0.796

holiday [10] and the price rise he attributed to MBG after schools reopened [11] are both consistent with this mechanism. The June 2026 rule that also stopped holiday meals for mothers and young children [28] deepened

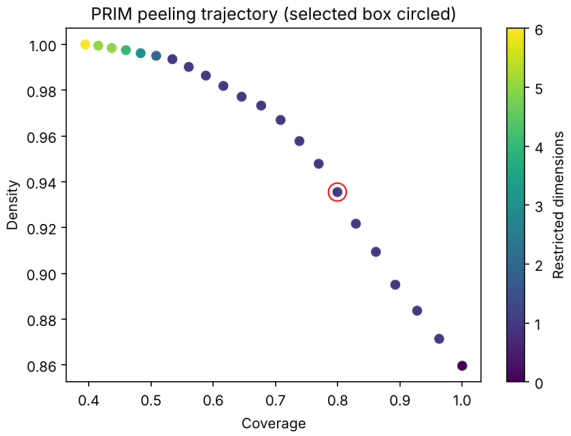


Figure 7: PRIM peeling trajectory (coverage versus density)

Figure 8: Decision-tree rules for spike occurrence (class 1 = spike)

```
|--- Chicken per beneficiary <= 0.14
|   |--- Egg servings after menu change <= 2.73
|   |   |--- Outbreak loss fraction <= 0.16
|   |   |   |--- weights: [233.00, 106.00] class: 0
|   |   |--- Outbreak loss fraction >  0.16
|   |   |   |--- weights: [112.00, 194.00] class: 1
|   |--- Egg servings after menu change >  2.73
|   |   |--- Feed shock magnitude <= 0.30
|   |   |   |--- weights: [215.00, 484.00] class: 1
|   |   |--- Feed shock magnitude >  0.30
|   |   |   |--- weights: [30.00, 420.00] class: 1
|--- Chicken per beneficiary >  0.14
|   |--- Chicken per beneficiary <= 0.18
|   |   |--- Household chicken elasticity <= -1.59
|   |   |   |--- weights: [71.00, 241.00] class: 1
|   |   |--- Household chicken elasticity >  -1.59
|   |   |   |--- weights: [29.00, 591.00] class: 1
|   |--- Chicken per beneficiary >  0.18
|   |   |--- Target kitchens <= 24028.67
|   |   |   |--- weights: [2.00, 8.00] class: 1
|   |   |--- Target kitchens >  24028.67
|   |   |   |--- weights: [10.00, 2254.00] class: 1
```

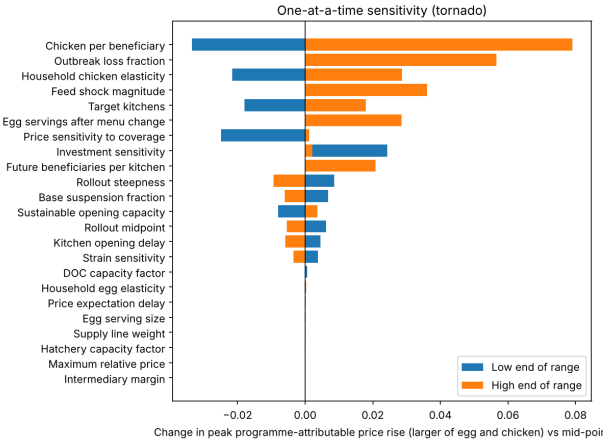


Figure 9: One-at-a-time sensitivity of peak programme-attributable price rise

these swings.

The broiler chain behaves differently. The calibrated sensitivity of broiler placements to the expected margin is close to zero, meaning that placements follow demand and integrator planning rather than prices. This is consistent with the integrator-dominated structure and quota-based allocation of day-old chicks described by Haryana et al. [17], and it explains why chicken price risk depends so

Table 11: Policy outcomes across scenarios (price effects are programme-attributable relative rises)

Policy	Spike prob.	95% CI	Peak egg (median)	Peak egg (P90)	Peak chicken (median)	Household egg loss	Cost per portion (Rp)	Coverage, final year
P0	0.860	0.850–0.869	0.118	0.246	0.119	0.0174	2,404	0.93
P1	0.851	0.841–0.861	0.128	0.277	0.111	0.0179	2,405	0.93
P2	0.860	0.850–0.869	0.118	0.246	0.119	0.0174	2,202	0.93
P3	0.657	0.643–0.670	0.121	0.263	-0.002	0.0156	2,255	0.93
P4	0.783	0.771–0.794	0.094	0.141	0.119	0.0151	2,309	0.93
P5	0.385	0.372–0.399	0.086	0.136	-0.002	0.0130	1,973	0.93

Figure 10: Early-warning decision-tree rules

```
|--- obs_chicken_coverage <= 0.99
|   |--- obs_egg_coverage <= 1.00
|   |   |--- weights: [62.00, 383.00] class: 1
|   |--- obs_egg_coverage >  1.00
|   |   |--- weights: [40.00, 950.00] class: 1
|--- obs_chicken_coverage >  0.99
|   |--- obs_egg_coverage <= 1.00
|   |   |--- weights: [202.00, 532.00] class: 1
|   |--- obs_egg_coverage >  1.00
|   |   |--- weights: [398.00, 2433.00] class: 1
```

Table 12: Paired comparison with P0 (Wilcoxon signed-rank)

Outcome	Policy	Median diff.	Rank-biserial r	Holm p
Peak egg price	P1	0.0073	0.508	<0.001
Peak egg price	P2	0.0000	0.000	1.000
Peak egg price	P3	0.0044	0.160	<0.001
Peak egg price	P4	-0.0196	-0.989	<0.001
Peak egg price	P5	-0.0297	-0.957	<0.001
Peak chicken price	P1	-0.0052	-0.998	<0.001
Peak chicken price	P2	0.0000	0.000	1.000
Peak chicken price	P3	-0.0992	-1.000	<0.001
Peak chicken price	P4	0.0000	-0.998	<0.001
Peak chicken price	P5	-0.0994	-1.000	<0.001
Household egg loss	P1	0.0002	0.592	<0.001
Household egg loss	P2	0.0000	0.000	1.000
Household egg loss	P3	-0.0011	-0.413	<0.001
Household egg loss	P4	-0.0020	-0.768	<0.001
Household egg loss	P5	-0.0045	-0.909	<0.001
Cost per portion	P1	-1.6	-0.316	<0.001
Cost per portion	P2	-195.7	-1.000	<0.001
Cost per portion	P3	-120.6	-1.000	<0.001
Cost per portion	P4	-83.2	-1.000	<0.001
Cost per portion	P5	-408.9	-1.000	<0.001

Table 13: Change in spike occurrence relative to P0 (McNemar)

Policy	Spikes removed	Spikes added	p (exact)
P1	161	120	0.017
P2	0	0	1.000
P3	1,137	122	<0.001
P4	383	0	<0.001
P5	2,381	9	<0.001

Table 14: Spike probability under alternative spike definitions (peak programme-attributable rise above a threshold, or above 10 points for at least 4 or 13 weeks)

Policy	Peak >5%	Peak >10%	Peak >15%	Peak >20%	≥4 wk	≥13 wk
P0	0.999	0.860	0.518	0.232	0.805	0.687
P1	0.997	0.851	0.513	0.255	0.799	0.674
P2	0.999	0.860	0.518	0.232	0.805	0.687
P3	0.967	0.657	0.364	0.196	0.605	0.469
P4	0.998	0.783	0.333	0.038	0.711	0.575
P5	0.945	0.385	0.078	0.009	0.334	0.232

strongly on how much chicken the programme buys.

6.2 Comparison with Previous Studies and the Attribution Debate

The counterfactual design speaks directly to the disagreement between BPS and the Ministry of Agriculture. In October 2025 the seasonally adjusted national egg price was 9.4% above its 2024 level and the chicken price 9.5%. In the calibrated baseline, the programme accounts for 5.0 percentage points of the egg price and 5.2 points of the chicken price in that month. The programme therefore explains a substantial part of the October 2025 egg price rise, supporting the BPS interpretation [6], while seasonal factors explain the rest. Neither agency's position can be confirmed from price data alone, because programme demand and seasonal demand rose at the same time; separating them requires a counterfactual.

The results agree with Mitchell, Thompson and Malone [16] that biological lags determine how far prices move after a shock: in the model the size of the programme-attributable rise depends on how fast demand arrives relative to the rearing delay. They extend Haryana et al. [17], who show that broiler price instability arises from the structure of supply, by adding a public programme as a new source of demand-side instability. The results are also consistent with the argument of Gelli et al. [13] that the market effects of school meals depend on procurement scale: MBG procurement is large relative to the national egg market, which is why national prices respond. Local effects in less integrated markets may be larger than the national effects estimated here.

6.3 Practical Implications for Programme Design

Four implications follow. First, procurement design matters more than roll-out speed. Forward contracts, which let farmers expand flocks before programme demand arrives, reduce spike probability from 86.0% to 65.7%, while slowing the roll-out (P1) reduces it only to 85.1% and creates spikes in 120 scenarios where none occurred before, because a slower roll-out can shift demand into a less favourable phase of the supply cycle. BGN could publish roll-out schedules and menu plans several months ahead and contract a share of eggs and chicken with farmer cooperatives on that basis.

Second, menus should flex by rule rather than by ad hoc decision. Menu flexibility (P4) lowers spike probability to 78.3% and cuts the 90th percentile of the egg price rise from 24.6 to 14.1 percentage points. The November 2025

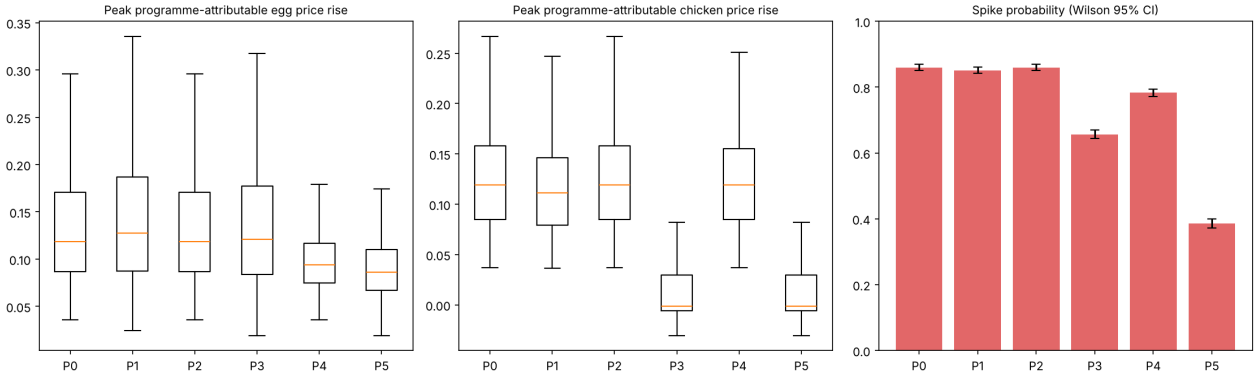


Figure 11: Policy comparison across scenarios

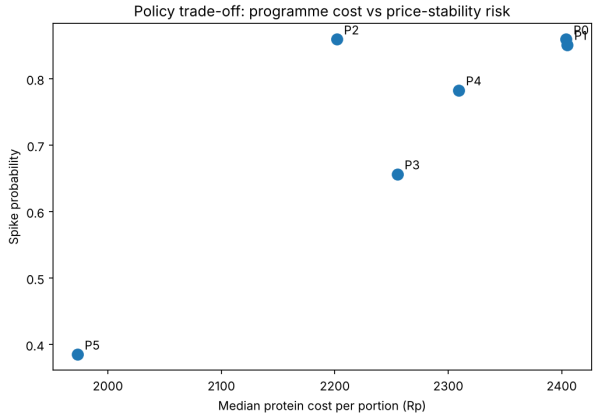


Figure 12: Protein cost per portion versus spike probability

and June 2026 menu changes show that BGN already uses menus as a market instrument [8,9]; a published rule linking servings to price thresholds would make the response predictable for farmers.

Third, cooperative sourcing (P2) lowers the protein cost per portion from Rp 2,404 to Rp 2,202 but does not change prices in the model, because it changes who captures the margin, not how much is produced. It is worth pursuing for budget reasons, and its price effect would depend on whether cooperatives also expand supply. Combining the three procurement measures (P5) gives the lowest spike probability (38.5%) and the lowest cost per portion.

Fourth, the programme should plan for school holidays. Keeping meals for mothers and young children during holidays, aligning holiday periods with government purchasing for food reserves, or shifting part of holiday-period egg supply to processing would reduce the stop-start shock that the June 2026 rules amplified. Because early market indicators predict spikes only moderately, these measures need to be designed in advance rather than triggered by price signals.

6.4 Methodological Contribution

The analysis shows what the combination of system dynamics and scenario discovery adds to a single simulation or a one-at-a-time sensitivity analysis. The counterfactual design separates the programme's effect from feed, disease and seasonal shocks that would raise prices anyway, which is the question the agencies disagree on. A one-at-a-time analysis finds the same main driver, but it describes behaviour around a single base case; only the ensemble gives the probability of a spike across the uncertainty space and shows how policies change it. PRIM turns the ensemble into conditions that decision makers can act on, such as the chicken quantity per beneficiary beyond which spikes become likely, and SHAP values show the direction and size of each driver's effect.

7 Limitations and Future Work

The study has several limitations. First, the model is national and has no regional detail or trade between regions; local price effects, particularly in eastern Indonesia, may differ from the national effects estimated here. Second, the chicken quantity per beneficiary is not published and official estimates conflict, so it is treated as a wide uncertainty; because it is a leading driver of spikes, publication of menu specifications by BGN would sharpen the results. Third, the temporary substitution of eggs requested in November 2025 is not modelled, because its size and duration were not reported, and the school-holiday calendar is a national approximation. Fourth, cooperative sourcing affects only procurement cost in the model. Fifth, household price elasticities come from a provincial study and are applied nationally, and the kitchen suspension hazard is constant although suspensions occurred in enforcement waves.

Sixth, the model reproduces the main movements of seasonally adjusted prices with moderate accuracy (R^2 of 0.45 for eggs and 0.57 for chicken); short-term fluctuations driven by events outside the model, such as weather, disease outbreaks and government market operations, are not reproduced. Two parameters, the price-expectation delay and the price response to inventory coverage, are not identifiable from the available data and were explored over wide ranges instead.

Future work could add regional subscripts and inter-island trade, collect procurement data from a sample of SPPG to replace the menu uncertainty with observed quantities, model cooperative sourcing as a channel for

supply expansion, apply behaviour-based scenario discovery [22] to classify the shape of price paths rather than only their peaks, and extend the model to other programme commodities such as milk, rice and vegetables.

8 Conclusion

This paper developed a system dynamics model of Indonesia's egg and broiler supply chains under the Free Nutritious Meals programme and analysed it with machine-learning scenario discovery over 5,000 scenarios. On RQ1, the programme's price-inelastic demand meets a biologically delayed supply response, so prices rise until households cut consumption and new flocks arrive; in the calibrated baseline the programme-attributable rise peaks at 11.2 percentage points for eggs and 12.8 for chicken, and school holidays add recurring demand swings. On RQ2, a spike of more than 10 points occurs in 86.0% of scenarios under current policy and depends mainly on chicken quantity per beneficiary, egg servings per week after June 2026 and size of a disease outbreak; spikes occur in 94% of scenarios with chicken quantity per beneficiary above 0.128 kg per week. On RQ3, forward contracting is the most effective single policy (65.7%), and combining cooperative sourcing, forward contracts and menu flexibility reduces spike probability to 38.5% at a lower cost per portion, without reducing programme coverage.

The contributions are a calibrated dynamic model of programme-driven demand shocks, a counterfactual design that separates programme effects from seasonal and external shocks, and an ensemble analysis that turns thousands of simulations into spike probabilities, thresholds and paired policy comparisons that a single run or a one-at-a-time analysis cannot provide. The results suggest that large public food programmes should plan procurement, menus and holiday supply in advance, because by the time price signals appear the supply response is months away. Extending the model to regional markets and other commodities is the natural next step.

Declarations

Funding: This research received no external funding.

Competing interests: The author declares no competing interests.

Data and code availability: The Vensim model, the Python analysis code, the price extract and the programme data with their sources are available from the author's repository (github.com/daniel766hi/kaggle-compe, folder paper/).

Use of AI tools: An AI assistant (Claude, Anthropic) was used to help develop the simulation and analysis code, collect and process public data, and draft and edit the text. The author reviewed, revised and verified all content, including every equation, data source and reference, and takes full responsibility for the work.

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A Model Equations

Variable	Equation	Units
INITIAL TIME	0	Week
FINAL TIME	312	Week
TIME STEP	0.03125	Week [0,?]
SAVEPER	1	Week [0,?]
Programme start time	53	Week
Target kitchens	29767	kitchen
Rollout midpoint	92.05	Week
Rollout steepness	7.39	Week
Rollout fraction	IF THEN ELSE(Time >= Programme start time, 1 / (1 + EXP(-(Time - Rollout midpoint) / Rollout steepness)), 0)	Dmnl
Planned kitchens	Target kitchens * Rollout fraction	kitchen
Operational kitchens	INTEG(Kitchen opening rate + Reinstatement rate - Suspension rate, 0)	kitchen
Suspended kitchens	INTEG(Suspension rate - Reinstatement rate, 0)	kitchen
Kitchen opening delay	4	Week
Kitchen opening rate	MAX(0, Planned kitchens - Operational kitchens - Suspended kitchens) / Kitchen opening delay	kitchen/Week
Reinstatement time	14.96	Week
Reinstatement rate	Suspended kitchens / Reinstatement time	kitchen/Week
Base suspension fraction	0.00409	1/Week
Sustainable opening capacity	300	kitchen/Week
Scaling strain	Kitchen opening rate / Sustainable opening capacity	Dmnl
Strain sensitivity	2	Dmnl
Effect of scaling strain on suspension	1 + Strain sensitivity * MAX(0, Scaling strain - 1)	Dmnl
Suspension rate	Operational kitchens * Base suspension fraction * Effect of scaling strain on suspension	kitchen/Week
Beneficiaries per kitchen	IF THEN ELSE(Time <= Last observed beneficiary week, Observed beneficiaries per kitchen(Time), Observed beneficiaries per kitchen(Last observed beneficiary week) + (Future beneficiaries per kitchen - Observed beneficiaries per kitchen(Last observed beneficiary week)) * MIN(1, (Time - Last observed beneficiary week) / Beneficiary adjustment time))	person/kitchen
Last observed beneficiary week	143.29	Week
Future beneficiaries per kitchen	2000	person/kitchen
Beneficiary adjustment time	26	Week
Beneficiaries served	Operational kitchens * Beneficiaries per kitchen	person
Holiday service share	IF THEN ELSE(Time >= Holiday rule change time, 0, Holiday service share before rule change)	Dmnl
Holiday service share before rule change	0.2	Dmnl
Holiday rule change time	128.625	Week
Programme in session	1 - School holiday indicator(Time) * (1 - Holiday service share)	Dmnl
Programme coverage	ZIDZ(Beneficiaries served, Target kitchens * Beneficiaries per kitchen)	Dmnl
Portions per beneficiary	5	portion/(person*Week)
Egg per beneficiary	Egg serving size * IF THEN ELSE(Time >= Egg menu change time, Egg servings after menu change, Egg servings before menu change)	kg/(person*Week)
Egg serving size	0.06	kg/serving
Egg servings before menu change	2	serving/(person*Week)
Egg servings after menu change	3	serving/(person*Week)
Egg menu change time	127.0	Week
Chicken per beneficiary	0.18	kg/(person*Week)
Menu flexibility switch	0	Dmnl
Max menu substitution	0.5	Dmnl
Substitution price threshold	1.15	Dmnl
Egg menu multiplier	1 - Menu flexibility switch * Max menu substitution * MIN(1, MAX(0, (Relative egg price - Substitution price threshold) / 0.2))	Dmnl
Chicken menu multiplier	1 - Menu flexibility switch * Max menu substitution * MIN(1, MAX(0, (Relative chicken price - Substitution price threshold) / 0.2))	Dmnl
MBG egg demand	Beneficiaries served * Egg per beneficiary * Egg menu multiplier * Programme in session / 1000	t/Week
MBG chicken demand	Beneficiaries served * Chicken per beneficiary * Chicken menu multiplier * Programme in session / 1000	t/Week
Forward contract share	0	Dmnl
Contract lead time	20	Week
Anticipated rollout fraction	IF THEN ELSE(Time + Contract lead time >= Programme start time, 1 / (1 + EXP(-(Time + Contract lead time - Rollout midpoint) / Rollout steepness)), 0)	Dmnl
Anticipated MBG egg demand	Target kitchens * Anticipated rollout fraction * Beneficiaries per kitchen * Egg per beneficiary / 1000	t/Week
Anticipated MBG chicken demand	Target kitchens * Anticipated rollout fraction * Beneficiaries per kitchen * Chicken per beneficiary / 1000	t/Week
Cooperative sourcing share	0.1	Dmnl
Intermediary margin	0.2	Dmnl
Egg procurement price	Egg price * (1 + Intermediary margin * (1 - Cooperative sourcing share))	Rp/kg
Chicken procurement price	Chicken price * (1 + Intermediary margin * (1 - Cooperative sourcing share))	Rp/kg
Programme protein cost	(MBG egg sales * Egg procurement price + MBG chicken sales * Chicken procurement price) * 1000	Rp/Week

Variable	Equation	Units
Protein cost per portion	Programme protein cost / MAX(1, Beneficiaries served * Portions per beneficiary)	Rp/portion
Feed shock magnitude	0	Dmnl
Feed shock time	120	Week
Feed shock duration	26	Week
Exogenous feed price index	1 + Feed shock magnitude * PULSE(Feed shock time, Feed shock duration)	Dmnl
Broiler feed weight	0.5	Dmnl
Total birds	Laying hens + Pullets + Broilers in growout * Broiler feed weight	bird
Initial total birds	INITIAL(Total birds)	bird
Feed price sensitivity to flock	0.3	Dmnl
Relative feed price	Exogenous feed price index * (Total birds / Initial total birds) ^ Feed price sensitivity to flock	Dmnl
Egg mass	0.06	kg/egg
Lay rate	5.9	egg/(bird*Week)
Production per hen	Lay rate * Egg mass / 1000	t/(bird*Week)
Initial egg production	121640	t/Week
Initial laying hens	Initial egg production / Production per hen	bird
Productive life	65	Week
Rearing time	18	Week
Laying hens	INTEG(Maturation - Culling - Disease loss, Initial laying hens)	bird
Pullets	INTEG(Pullet placement - Maturation, Initial laying hens / Productive life * Rearing time)	bird
Maturation	DELAY3(Pullet placement, Rearing time)	bird/Week
Culling	Laying hens / Productive life	bird/Week
Outbreak loss fraction	0	Dmnl
Outbreak time	150	Week
Outbreak duration	4	Week
Disease loss	Laying hens * (-LN(1 - MIN(Outbreak loss fraction, 0.999))) * PULSE(Outbreak time, Outbreak duration) / Outbreak duration	bird/Week
Egg production	Laying hens * Production per hen	t/Week
Egg spoilage fraction	0.01	1/Week
Normal egg coverage	2	Week
Reference household egg demand	Initial egg production / (1 + Egg spoilage fraction * Normal egg coverage)	t/Week
Egg inventory	INTEG(Egg production - Egg sales - Egg spoilage, Reference household egg demand * Normal egg coverage)	t
Egg spoilage	Egg inventory * Egg spoilage fraction	t/Week
Minimum delivery time	0.5	Week
Egg available	Egg inventory / Minimum delivery time	t/Week
Household egg elasticity	-0.669	Dmnl
Household egg demand	Reference household egg demand * Relative egg price ^ Household egg elasticity	t/Week
MBG egg sales	MIN(MBG egg demand, Egg available)	t/Week
Household egg sales	MIN(Household egg demand, Egg available - MBG egg sales)	t/Week
Egg sales	MBG egg sales + Household egg sales	t/Week
Total egg demand	MBG egg demand + Household egg demand	t/Week
Relative egg coverage	(Egg inventory / MAX(1, Total egg demand)) / Normal egg coverage	Dmnl
Baseline egg price	28140	Rp/kg
Normal egg margin	0.1	Dmnl
Egg feed cost share	0.7	Dmnl
Base egg unit cost	Baseline egg price / (1 + Normal egg margin)	Rp/kg
Egg unit cost	Base egg unit cost * (Egg feed cost share * Relative feed price + 1 - Egg feed cost share)	Rp/kg
Price sensitivity to coverage	1	Dmnl
Cost pull weight	0.3	Dmnl
Maximum relative price	3	Dmnl
Price adjustment time	2	Week
Indicated egg price	MIN(Maximum relative price * Baseline egg price, Egg price * Relative egg coverage ^ (-Price sensitivity to coverage) * (Egg unit cost * (1 + Normal egg margin) / Egg price) ^ Cost pull weight)	Rp/kg
Change in egg price	(Indicated egg price - Egg price) / Price adjustment time	Rp/(kg*Week)
Egg price	INTEG(Change in egg price, Baseline egg price)	Rp/kg
Relative egg price	Egg price / Baseline egg price	Dmnl
Price expectation delay	8	Week
Expected egg price	SMOOTH(Egg price, Price expectation delay)	Rp/kg
Expected egg margin	(Expected egg price - Egg unit cost) / Egg unit cost	Dmnl
Egg farmer margin	(Egg price - Egg unit cost) / Egg unit cost	Dmnl
Investment sensitivity	1.5	Dmnl
Effect of margin on desired hens	MAX(0.5, 1 + Investment sensitivity * (Expected egg margin - Normal egg margin))	Dmnl
Desired laying hens	Laying hens * Effect of margin on desired hens + Forward contract share * MAX(0, Anticipated MBG egg demand - MBG egg demand) / Production per hen	bird
Expected hen loss	SMOOTH(Culling + Disease loss, 4)	bird/Week
Desired pullets	Expected hen loss * Rearing time	bird
Capacity adjustment time	26	Week
Supply line weight	0.5	Dmnl
Hatchery capacity factor	1.5	Dmnl
Max pullet placement	Hatchery capacity factor * Initial laying hens / Productive life	bird/Week
Indicated pullet placement	Expected hen loss + (Desired laying hens - Laying hens) / Capacity adjustment time + Supply line weight * (Desired pullets - Pullets) / Capacity adjustment time	bird/Week

Variable	Equation	Units
Pullet placement	MIN(Max pullet placement, MAX(0, Indicated pullet placement))	bird/Week
Household egg consumption ratio	Household egg sales / Reference household egg demand	Dmnl
Initial chicken production	73565	t/Week
Meat per bird	1.37	kg/bird
Growout time	5	Week
Growout mortality	0.05	Dmnl
Initial DOC placement	Initial chicken production * 1000 / Meat per bird / (1 - Growout mortality)	bird/Week
Broilers in growout	INTEG(DOC placement - Broiler harvest - Growout mortality loss, Initial DOC placement * Growout time)	bird
Broiler harvest	Broilers in growout / Growout time * (1 - Growout mortality)	bird/Week
Growout mortality loss	Broilers in growout / Growout time * Growout mortality	bird/Week
Chicken production	Broiler harvest * Meat per bird / 1000	t/Week
Chicken spoilage fraction	0.02	1/Week
Normal chicken coverage	1	Week
Reference household chicken demand	Initial chicken production / (1 + Chicken spoilage fraction * Normal chicken coverage)	t/Week
Chicken inventory	INTEG(Chicken production - Chicken sales - Chicken spoilage, Reference household chicken demand * Normal chicken coverage)	t
Chicken spoilage	Chicken inventory * Chicken spoilage fraction	t/Week
Chicken available	Chicken inventory / Minimum delivery time	t/Week
Household chicken elasticity	-1.581	Dmnl
Household chicken demand	Reference household chicken demand * Relative chicken price ^ Household chicken elasticity	t/Week
MBG chicken sales	MIN(MBG chicken demand, Chicken available)	t/Week
Household chicken sales	MIN(Household chicken demand, Chicken available - MBG chicken sales)	t/Week
Chicken sales	MBG chicken sales + Household chicken sales	t/Week
Total chicken demand	MBG chicken demand + Household chicken demand	t/Week
Relative chicken coverage	(Chicken inventory / MAX(1, Total chicken demand)) / Normal chicken coverage	Dmnl
Baseline chicken price	35704	Rp/kg
Normal chicken margin	0.1	Dmnl
Chicken feed cost share	0.65	Dmnl
Base chicken unit cost	Baseline chicken price / (1 + Normal chicken margin)	Rp/kg
Chicken unit cost	Base chicken unit cost * (Chicken feed cost share * Relative feed price + 1 - Chicken feed cost share)	Rp/kg
Indicated chicken price	MIN(Maximum relative price * Baseline chicken price, Chicken price * Relative chicken coverage ^ (-Price sensitivity to coverage) * (Chicken unit cost * (1 + Normal chicken margin) / Chicken price) ^ Cost pull weight)	Rp/kg
Change in chicken price	(Indicated chicken price - Chicken price) / Price adjustment time	Rp/(kg*Week)
Chicken price	INTEG(Change in chicken price, Baseline chicken price)	Rp/kg
Relative chicken price	Chicken price / Baseline chicken price	Dmnl
Expected chicken price	SMOOTH(Chicken price, Price expectation delay / 2)	Rp/kg
Expected chicken margin	(Expected chicken price - Chicken unit cost) / Chicken unit cost	Dmnl
Broiler investment sensitivity	1	Dmnl
Effect of margin on broiler placement	MAX(0.5, 1 + Broiler investment sensitivity * (Expected chicken margin - Normal chicken margin))	Dmnl
Demand forecast time	4	Week
Expected chicken demand	SMOOTH(Total chicken demand, Demand forecast time)	t/Week
Inventory adjustment time	2	Week
Desired chicken production	(Expected chicken demand + Chicken spoilage) * Effect of margin on broiler placement + (Expected chicken demand * Normal chicken coverage - Chicken inventory) / Inventory adjustment time + Forward contract share * MAX(0, Anticipated MBG chicken demand - MBG chicken demand)	t/Week
DOC capacity factor	1.2	Dmnl
Max DOC placement	DOC capacity factor * Initial DOC placement	bird/Week
DOC placement	MIN(Max DOC placement, MAX(0, Desired chicken production * 1000 / Meat per bird / (1 - Growout mortality)))	bird/Week
Household chicken consumption ratio	Household chicken sales / Reference household chicken demand	Dmnl

B Parameters and Sources

Parameter	Value	Units	Source or basis
INITIAL TIME	0.0	Week	Week 0 = first week of Jan 2024.
FINAL TIME	312.0	Week	Six years: 2024-2029.
TIME STEP	0.03125	Week [0,?]	Halved in integration tests B1/B2. The shortest time constant is the 0.5-week minimum delivery time, which binds when demand jumps as schools reopen; 0.0625 failed B1 (1.5 percent of range, chicken price after school holidays), so the step was halved to 0.03125 (1/16 of that time constant).
SAVEPER	1.0	Week [0,?]	
Programme start time	53.0	Week	MBG launched 6 January 2025 (week 53 from 1 Jan 2024).
Target kitchens	29767.0	kitchen	From data: fitted so that modelled operational SPPG match BGN counts (fit_rollout.py, data/rollout_fit.csv).
Rollout midpoint	92.05	Week	From data: fitted to BGN SPPG counts via operational kitchens (fit_rollout.py).

Parameter	Value	Units	Source or basis
Rollout steepness	7.39	Week	From data: fitted to BGN SPPG counts via operational kitchens (fit_rollout.py). POLICY P1 (phased roll-out) increases it.
Kitchen opening delay	4.0	Week	ASSUMPTION (no public data): weeks from planning to operation. Explored 2-8 in the ensemble. UNCERTAINTY.
Reinstatement time	14.96	Week	From data: fitted with Base suspension fraction (fit_suspensions.py).
Base suspension fraction	0.00409	1/Week	From data: fitted to BGN reports of 8,182 SPPG ever suspended and 2,213 inactive by end May 2026 (fit_suspensions.py). UNCERTAINTY.
Sustainable opening capacity	300.0	kitchen/Week	ASSUMPTION (no public data): openings per week the system can supervise without quality loss. Observed openings peaked at approx. 700 per week in late 2025. Explored 150-700 in the ensemble. UNCERTAINTY.
Strain sensitivity	2.0	Dmnl	UNCERTAINTY: strength of the scaling-strain effect on suspensions.
Last observed beneficiary week	143.29	Week	30 Sep 2026.
Future beneficiaries per kitchen	2000.0	person/kitchen	UNCERTAINTY (1,900-2,900): from the Sep 2026 level up to the earlier level; BGN cap 2,500 portions per SPPG (3,000 with certified cooks), Keputusan Kepala BGN 244/2025.
Beneficiary adjustment time	26.0	Week	Time to move from the last observed ratio to the future level.
Holiday service share before rule change	0.2	Dmnl	Pregnant and breastfeeding mothers and toddlers kept receiving meals in holidays (about 500 of 2,500 portions per SPPG, Keputusan Kepala BGN 244/2025); school deliveries in holidays were optional.
Holiday rule change time	128.625	Week	BGN circular SE 12/2026 (June 2026): distribution to all groups stops on school holidays, public holidays and weekends.
Portions per beneficiary	5.0	portion/(person*Week)	Meals are served on school days only (five per week) following the 2026 schedule change; previously six. Affects cost per portion only.
Egg serving size	0.06	kg/serving	One egg, approx. 60 g. UNCERTAINTY 0.05-0.065.
Egg servings before menu change	2.0	serving/(person*Week)	BGN planning assumption (two egg dishes a week).
Egg servings after menu change	3.0	serving/(person*Week)	At least three a week from 8 June 2026 (described as temporary). UNCERTAINTY 2-4.
Egg menu change time	127.0	Week	8 June 2026.
Chicken per beneficiary	0.18	kg/(person*Week)	No published menu specification. Official estimates conflict: Agriculture Minister 1.1 Mt/yr for the full programme (approx. 0.255 kg); Bapanas SPPG need 222.7 kt in 2025; another report 70 kt/yr. Treated as deep UNCERTAINTY 0.08-0.26.
Menu flexibility switch	0.0	Dmnl	POLICY P4 (0/1): substitute protein when price is high.
Max menu substitution	0.5	Dmnl	Max share of egg/chicken replaced (fish, tempeh, tofu).
Substitution price threshold	1.15	Dmnl	Relative price above which substitution starts.
Forward contract share	0.0	Dmnl	POLICY P3 (0-1): share of future programme demand contracted in advance with farmers.
Contract lead time	20.0	Week	How far ahead contracted demand is announced.
Cooperative sourcing share	0.1	Dmnl	POLICY P2 (0-1): share bought directly from farmer groups/cooperatives.
Intermediary margin	0.2	Dmnl	Mark-up paid by SPPG buying via traders relative to market price. Evidence: BPS trade and transport margin (MPP) for eggs 26.8% (2017), 13.1% (2018), 20.2% (2021). Range 0.10-0.30.
Feed shock magnitude	0.0	Dmnl	UNCERTAINTY: relative rise in world feed price during the shock.
Feed shock time	120.0	Week	
Feed shock duration	26.0	Week	
Broiler feed weight	0.5	Dmnl	Feed use of a broiler relative to a layer.
Feed price sensitivity to flock	0.3	Dmnl	Loop B3 strength.
Egg mass	0.06	kg/egg	Standard chicken egg approx. 55-62 g (food exchange list: 55 g; commercial approx. 60 g).
Lay rate	5.9	egg/(bird*Week)	Average hen-day production approx. 84.5% (Lohmann Brown, Indonesian field study) = 5.9 eggs per hen per week. Scales flock size only.
Initial egg production	121640.0	t/Week	BPS 2024 (preliminary): 6,342,705 t laying-hen eggs per year / 52.14 weeks.
Productive life	65.0	Week	Laying from approx. week 19 to culling at approx. 80-90 weeks of age (Indonesian layer practice; some flocks to 100+). Range 60-85.
Rearing time	18.0	Week	Pullets reach point of lay at approx. 16-21 weeks (Indonesian layer studies; Lohmann/Hy-Line guides). Range 16-21.
Outbreak loss fraction	0.0	Dmnl	Scenario input. Avian influenza is endemic in Indonesia; large flock losses are uncommon. Range 0-0.3 explored as stress test.
Outbreak time	150.0	Week	
Outbreak duration	4.0	Week	
Egg spoilage fraction	0.01	1/Week	
Normal egg coverage	2.0	Week	Normal inventory in the distribution chain, in weeks of demand.
Minimum delivery time	0.5	Week	
Household egg elasticity	-0.669	Dmnl	Own-price elasticity of eggs, QUAIDS on SUSENAS 2021 (Khoiriyah et al. 2024). Ensemble range -1.0 to -0.5 (Bengkulu 2021: -0.82).
Baseline egg price	28140.0	Rp/kg	From data: 2024 mean of monthly national median retail price, WFP/HDX Indonesia Food Prices (see data/raw/SOURCE.txt).
Normal egg margin	0.1	Dmnl	
Egg feed cost share	0.7	Dmnl	Feed approx. 60-70% of poultry production cost (Kementan); 70% often cited for layers. Range 0.6-0.8.
Price sensitivity to coverage	1.0	Dmnl	Not identified by the price data (fit error nearly flat between 1 and 10; s02 identifiability profile). Fixed at 1 and explored 0.5-3 in the ensemble. UNCERTAINTY.
Cost pull weight	0.3	Dmnl	How strongly price is pulled toward cost plus normal margin.
Maximum relative price	3.0	Dmnl	UNCERTAINTY: price ceiling relative to baseline set by imports, substitutes and government market operations.
Price adjustment time	2.0	Week	
Price expectation delay	8.0	Week	UNCERTAINTY: farmer expectation-formation delay. Not identified by the price data (calibration moved it between 2 and 30 weeks); fixed at 8 weeks and explored over 4-16 in the ensemble. Find a literature value.
Investment sensitivity	1.5	Dmnl	UNCERTAINTY: response of desired flock to expected margin (loop B1).
Capacity adjustment time	26.0	Week	
Supply line weight	0.5	Dmnl	UNCERTAINTY: 1 = farmers fully account for pullets in rearing; <1 causes overshoot.

Parameter	Value	Units	Source or basis
Hatchery capacity factor	1.5	Dmnl	UNCERTAINTY: max pullet placement relative to baseline replacement.
Initial chicken production	73565.0	t/Week	BPS 2024 (preliminary): 3,835,917 t broiler meat per year / 52.14 weeks.
Meat per bird	1.37	kg/bird	Harvest at 30-36 days, live weight approx. 1.7-2.2 kg (Indonesian farm reports), carcass approx. 70-75% of live weight: approx. 1.9 kg x 0.72. Scales bird numbers only.
Growout time	5.0	Week	
Growout mortality	0.05	Dmnl	Indonesian farm reports 1.5-5.1%; under 4% considered good.
Chicken spoilage fraction	0.02	1/Week	
Normal chicken coverage	1.0	Week	
Household chicken elasticity	-1.581	Dmnl	Own-price elasticity of poultry, QUAIDS on SUSENAS 2021 (Khoiriyah et al. 2024). Ensemble range -1.8 to -1.2 (Bengkulu 2021: -1.54).
Baseline chicken price	35704.0	Rp/kg	From data: 2024 mean of monthly national median retail price, WFP/HDX Indonesia Food Prices (see data/raw/SOURCE.txt).
Normal chicken margin	0.1	Dmnl	
Chicken feed cost share	0.65	Dmnl	Feed approx. 60-70% of broiler production cost (Kementan; Indonesian farm studies).
Broiler investment sensitivity	1.0	Dmnl	UNCERTAINTY
Demand forecast time	4.0	Week	
Inventory adjustment time	2.0	Week	
DOC capacity factor	1.2	Dmnl	Day-old-chick supply chronically exceeds demand: Jan 2020 projected production 317 million vs demand 259.6 million (Kontan), ratio approx. 1.22; repeated government culling programmes. Range 1.05-1.4.